

Problem 1

a)
$$\begin{array}{r} 19 \overline{) 7} \\ -14 \overline{) 2} \rightarrow \text{quotient} \\ \hline 5 \text{ remainder} \end{array}$$

b)
$$\begin{array}{r} -111 \overline{) 11} \\ +12 \overline{) 10} \rightarrow -11 \text{ (9)} \\ \hline +10 \text{ (1)} \end{array}$$

c)
$$\begin{array}{r} 789 \overline{) 23} \\ -69 \overline{) 34} \\ \hline 99 \\ -92 \\ \hline 7 \end{array}$$

d)
$$\begin{array}{r} 1001 \overline{) 13} \\ -91 \overline{) 77} \\ \hline 91 \\ -91 \\ \hline 0 \end{array}$$

e)
$$\begin{array}{r} 0 \overline{) 19} \\ 0 \overline{) 0} \\ \hline 0 \end{array}$$

f)
$$\begin{array}{r} 3 \overline{) 5} \\ -0 \overline{) 0} \\ \hline 3 \end{array}$$

g)
$$\begin{array}{r} -1 \overline{) 3} \\ -3 \overline{) 0} \rightarrow -1 \\ \hline +2 \end{array}$$

h)
$$\begin{array}{r} 4 \overline{) 11} \\ -4 \overline{) 4} \\ \hline 0 \end{array}$$

Problem 2

For instance, $-3 \dots 2 \dots 7$

$a = 2, m = 5$

$r = a \% m = 2 \% 5 = 2$

$r \leq m/2 \ (2 \leq 2.5) \Rightarrow res = r = 2$

$$\text{result} = \begin{cases} r \rightarrow r \leq m/2 \\ r-m \rightarrow r > m/2 \end{cases}$$

+1 fix: $a = -111, m = +11$

$$\begin{array}{r} -111 \overline{) +11} \\ +12 \overline{) 11} \\ \hline 10 \end{array}$$

$r > m/2 \Rightarrow r-m = -1$
 $(10 > 11/2) \Rightarrow (10 - 11)$

Problem 3

a) $a=0, b=2, c=2, m=4$

$a \equiv b \pmod{4} \Rightarrow 0 \equiv 2 \pmod{4} \times$

$ac \equiv bc \pmod{4} \Rightarrow 0 \cdot 2 \equiv 2 \cdot 2$
 $0 \equiv 4 \pmod{4} \checkmark$

b) $a=2, b=5, c=1, d=4, m=3$

$a \equiv b \Rightarrow 2 \equiv 5 \pmod{3} \checkmark$

$c \equiv d \Rightarrow 1 \equiv 4 \pmod{3} \checkmark$

$a^c \equiv b^d \Rightarrow 2^1 \equiv 5^4 \pmod{3}$

$2 \equiv 625 \pmod{3} \times$
 $(2 \equiv 1)$

Problem 4

$$a \cdot a \equiv b \cdot b \pmod{m} \Rightarrow a^2 \equiv b^2 \pmod{m}$$

$$a \cdot a \cdot a \equiv b \cdot b \cdot b \pmod{m} \Rightarrow a^3 \equiv b^3 \pmod{m}$$

$$\vdots$$
$$a^k \equiv b^k \pmod{m}$$

for instance, let k be 1

$$a=3, b=8, m=5$$

$$a \equiv b \pmod{m} \Rightarrow 3 \equiv 8 \pmod{5} \Rightarrow 3 \stackrel{k=1}{=} 8 \stackrel{k=1}{=} 8 \pmod{5} \checkmark$$

Problem 5

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$$0 +_5 0 = 0 \pmod{5} = 0$$

$$0 +_5 1 = 1 \pmod{5} = 1$$

$$0 +_5 2 = 2 \pmod{5} = 2$$

$$0 +_5 3 = 3 \pmod{5} = 3$$

$$0 +_5 4 = 4 \pmod{5} = 4$$

$$1 +_5 0 = 1 \pmod{5} = 1$$

$$1 +_5 1 = 2 \pmod{5} = 2$$

$$1 +_5 2 = 3 \pmod{5} = 3$$

$$1 +_5 3 = 4 \pmod{5} = 4$$

$$1 +_5 4 = 5 \pmod{5} = 0$$

$\cdot_5$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

$$0 \cdot_5 0 = 0 \pmod{5} = 0$$

$$0 \cdot_5 1 = 0 \pmod{5} = 0$$

$$-||- \quad -||-$$

$$1 \cdot_5 0 = 0 \pmod{5} = 0$$

$$1 \cdot_5 1 = 1 \pmod{5} = 1$$

$$-||- \quad -||-$$

Problem 6

$$f(a) = a \text{ div } 3 \Rightarrow f(1) = 1 \text{ div } 3 = 0 \quad \text{X one-to-one}$$

$$f(a) = a \text{ div } \sqrt{3} \Rightarrow f(2) = 2 \text{ div } \sqrt{3} = 0 \quad \checkmark \text{ onto}$$

$$g(a) = a \bmod 3 \Rightarrow g(1) = 1 \bmod 3 = 1 \quad \text{1 x one-to-one}$$

$\Rightarrow g(10) = 10 \bmod 3 = 1$ x onto (only 0, 1, 2 as an output)

	one-to-one	onto
$f(a) = a \text{ div } d$	NO	YES
$g(a) = a \bmod d$	NO	NO

Problem 7

a)
$$\begin{array}{r} 231 \overline{) 2} \\ - 230 \overline{) 115} \overline{) 2} \\ \hline 1 \quad 114 \overline{) 57} \overline{) 2} \\ \quad \quad \quad 1 \quad 56 \overline{) 28} \overline{) 2} \\ \quad \quad \quad \quad \quad 1 \quad 28 \overline{) 14} \overline{) 2} \\ \quad \quad \quad \quad \quad \quad \quad 0 \quad 14 \overline{) 7} \overline{) 2} \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad 0 \quad 6 \overline{) 3} \overline{) 2} \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 1 \quad 2 \overline{) 1} \overline{) 2} \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 1 \quad 0 \overline{) 0} \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 1 \end{array}$$

$$\underline{11100111}$$

b) $4532 \div 2 = 2266$
 $2266 \div 2 = 1133$
 $1133 \div 2 = 566$
 $566 \div 2 = 283$
 $283 \div 2 = 141$
 $141 \div 2 = 70$
 $70 \div 2 = 35$
 $35 \div 2 = 17$
 $17 \div 2 = 8$
 $8 \div 2 = 4$
 $4 \div 2 = 2$
 $2 \div 2 = 1$
 $1 \div 2 = 0$

1000110110100

$$c) 97644 \mid 2$$

$$97644 \mid 48822 \mid 2$$

$$0 \quad 48822 \mid 24411 \mid 2$$

$$0 \quad 24410 \mid 12205 \mid 2$$

$$1 \quad 12204 \mid 6102 \mid 2$$

$$1 \quad 6102 \mid 3051 \mid 2$$

$$0 \quad 3050 \mid 1525 \mid 2$$

$$1 \quad 1524 \mid 762 \mid 2$$

$$1 \quad 762 \mid 381 \mid 2$$

$$0 \quad 380 \mid 190 \mid 2$$

$$1 \quad 190 \mid 95 \mid 2$$

$$0 \quad 94 \mid 47 \mid 2$$

$$1 \quad 46 \mid 23$$

1

$$23 \mid 2$$

$$22 \mid 11 \mid 2$$

$$1 \quad 10 \mid 5 \mid 2$$

$$1 \quad 4 \mid 2 \mid 2$$

$$1 \quad 2 \mid 1 \mid 2$$

$$0 \quad 0 \mid 0$$

1

$$10111101011000$$

Problem 8

$$a) (11111)_2$$

$$16+8+4+2+1 = 31$$

$$b) (100000001)_2$$

$$512 + \dots + 1 = 513$$

$$c) (101010101)_2$$

$$256+0+64+0+16+0+4+0+1 = 341$$

$$d) (110100100010000)_2$$

$$2^{14} + 2^{13} + 2^{11} + 2^8 + 2^4$$

$$2^{14} + 2^{13} + 2^{11} + 2^8 + 2^4 = 26896$$

Problem 9

$$a) \begin{array}{r} 1000111 \\ + 1110111 \\ \hline 10111110 \end{array}$$

$$1000111$$

$$64 + \dots + 4 + 2 + 1 = 71$$

$$1110111$$

$$64 + 32 + 16 + \dots + 4 + 2 + 1 = 119$$

$$\begin{array}{r} 119 \\ \times 71 \\ \hline 119 \end{array}$$

$$\begin{array}{r} 833 \\ \times 8449 \\ \hline \end{array}$$

result

$$10000100000001$$

$$\begin{array}{r} 8449 \mid 2 \\ 8448 \mid 4224 \mid 2 \\ 1 \quad 4224 \mid 2112 \mid 2 \\ 0 \quad 2112 \mid 1056 \mid 2 \\ 0 \quad 1056 \mid 528 \mid 2 \\ 0 \quad 528 \mid 264 \mid 2 \\ 0 \quad 264 \mid 132 \mid 2 \\ 0 \quad 132 \mid 66 \mid 2 \\ 0 \quad 66 \mid 33 \mid 2 \\ 1 \quad 33 \mid 16 \mid 2 \\ 0 \quad 16 \mid 8 \mid 2 \\ 0 \quad 8 \mid 4 \mid 2 \\ 0 \quad 4 \mid 2 \mid 2 \\ 0 \quad 2 \mid 1 \mid 2 \\ 0 \quad 1 \mid 0 \end{array}$$

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 1 & 1 & 1 \\
 1 & 1 & 1 & 0 \\
 1 & 1 & 1 & 1 \\
 1 & 1 & 1 & 1
 \end{array} \\
 + \begin{array}{cccc}
 1 & 0 & 1 & 1 \\
 1 & 1 & 0 & 1 \\
 1 & 1 & 0 & 1 \\
 1 & 1 & 0 & 1
 \end{array} \\
 \hline
 \begin{array}{cccc}
 1 & 1 & 0 & 1 \\
 0 & 1 & 0 & 1 \\
 1 & 1 & 0 & 0 \\
 1 & 1 & 0 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{cccc}
 1 & 1 & 1 & 0 \\
 1 & 1 & 1 & 1
 \end{array}$$

$$128 + 64 + 32 + \dots + 8 + 4 + 2 + 1 = 239$$

$$\begin{array}{cccc}
 1 & 0 & 1 & 1 \\
 1 & 1 & 0 & 1
 \end{array}$$

$$128 + \dots + 32 + 16 + \dots + 8 + 4 + \dots + 1 = 189$$

$$\begin{array}{r}
 239 \\
 \times 189 \\
 \hline
 2151 \\
 + 1912 \\
 \hline
 45171
 \end{array}$$

$$\begin{array}{r}
 45171 \mid 2 \\
 - 45170 \mid 22585 \mid 2 \\
 \hline
 1 \mid 22584 \mid 11292 \mid 2 \\
 \hline
 1 \mid 11292 \mid 5646 \mid 2 \\
 \hline
 0 \mid 5646 \mid 2823 \mid 2 \\
 \hline
 0 \mid 2822 \mid 1411 \mid 2 \\
 \hline
 1 \mid 1410 \mid 705 \mid 2 \\
 \hline
 1 \mid 704 \mid 352 \mid 2 \\
 \hline
 1 \mid 352 \mid 176 \mid 2 \\
 \hline
 0 \mid 176 \mid 88 \mid 2 \\
 \hline
 0 \mid 88 \mid 44 \mid 2 \\
 \hline
 0 \mid 44 \mid 22 \mid 2 \\
 \hline
 0 \mid 22 \mid 11 \mid 2 \\
 \hline
 0 \mid 11 \mid 5 \mid 2 \\
 \hline
 1
 \end{array}$$

ans: 1011000001110011

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0
 \end{array} \\
 \hline
 \begin{array}{cccc}
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{cccc}
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0
 \end{array}$$

$$512 + 0 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 0 = 682$$

$$\begin{array}{cccc}
 1 & 1 & 1 & 1 \\
 1 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0
 \end{array}$$

$$256 + 128 + 64 + 32 + 16 + \dots = 496$$

$$\begin{array}{r}
 682 \\
 \times 496 \\
 \hline
 4092 \\
 6138 \\
 2728 \\
 \hline
 338272
 \end{array}$$

$$\begin{array}{r}
 338272 \mid 2 \\
 - 338272 \mid 169136 \mid 2 \\
 \hline
 0 \mid 169136 \mid 84568 \mid 2 \\
 \hline
 0 \mid 84568 \mid 42284 \mid 2 \\
 \hline
 0 \mid 42284 \mid 21142 \mid 2 \\
 \hline
 0 \mid 21142 \mid 10571 \mid 2 \\
 \hline
 1 \mid 10570 \mid 5285 \mid 2 \\
 \hline
 1 \mid 5284 \mid 2642 \mid 2 \\
 \hline
 0 \mid 2642 \mid 1321 \mid 2 \\
 \hline
 0 \mid 1320 \mid 660 \mid 2 \\
 \hline
 1 \mid 660 \mid 330 \mid 2 \\
 \hline
 0 \mid 330 \mid 165 \mid 2 \\
 \hline
 0
 \end{array}$$

ans: 10100101001011000000

$$\begin{array}{r} d) \quad 10 \ 0000 \ 0001 \\ + \quad 11 \ 1111 \ 1111 \\ \hline 110 \ 0000 \ 0000 \end{array}$$

$$\begin{array}{r} 10 \ 0000 \ 0001 \\ 1024 + \dots + 1 = 1025 \\ 512 \qquad \qquad \qquad 513 \end{array}$$

$$11 \ 1111 \ 1111$$

$$512 + 256 + 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 1023$$

$$\begin{array}{r} 1023 \\ \times 513 \\ \hline \end{array}$$

$$3069$$

$$1023$$

$$5115$$

$$524799$$

$$524799 \div 2$$

$$524798 \div 2 = 262399$$

$$262398 \div 2 = 131199$$

$$131198 \div 2 = 65599$$

$$65598 \div 2 = 32799$$

$$32798 \div 2 = 16399$$

$$16398 \div 2 = 8199$$

$$8198 \div 2 = 4099$$

$$4099 \div 2$$

$$4098 \div 2 = 2049$$

$$2048 \div 2 = 1024$$

$$1024 \div 2 = 512$$

$$512 \div 2 = 256$$

$$256 \div 2 = 128$$

$$128 \div 2 = 64$$

$$64 \div 2 = 32$$

$$32 \div 2 = 16$$

$$16 \div 2 = 8$$

$$8 \div 2 = 4$$

$$4 \div 2 = 2$$

$$2 \div 2 = 1$$

ans:

$$10 \ 0000 \ 0000 \ 01111111$$

Problem 10

$$\begin{array}{r} a) \quad 88 \div 2 \\ \hline 88 \div 2 = 44 \\ 44 \div 2 = 22 \\ 22 \div 2 = 11 \end{array}$$

$$2 \times 2 \times 2 \times 11$$

$$\begin{array}{r} b) \quad 126 \div 2 \\ \hline 126 \div 2 = 63 \\ 63 \div 3 = 21 \\ 21 \div 3 = 7 \end{array}$$

$$\begin{array}{r} 126 \div 2 \\ \hline 126 \div 2 = 63 \\ 63 \div 3 = 21 \\ 21 \div 3 = 7 \end{array}$$

$$2 \times 3 \times 3 \times 7$$

$$\begin{array}{r} c) \quad 729 \div 3 \\ \hline 729 \div 3 = 243 \\ 243 \div 3 = 81 \\ 81 \div 3 = 27 \\ 27 \div 3 = 9 \\ 9 \div 3 = 3 \\ 3 \div 3 = 1 \end{array}$$

$$3^6$$

$$\begin{array}{r} d) \quad 1001 \div 7 \\ \hline 1001 \div 7 = 143 \\ 143 \div 11 = 13 \end{array}$$

$$7 \times 11 \times 13$$

$$\begin{array}{r} e) \quad 1111 \div 11 \\ \hline 1111 \div 11 = 101 \end{array}$$

$$11 \times 101$$

